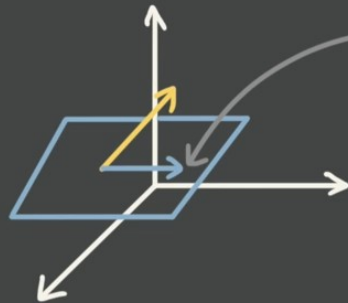


Orthonormal bases and Gram-Schmidt

Orthonormal set:

1. All vectors are normal
2. All vectors are orthogonal



Easier onto an
orthonormal subspace

Orthonormal bases

-orthogonal $\vec{v}_1 \cdot \vec{v}_2 = 0$

-Normal $\vec{v}_1 \cdot \vec{v}_1 = \|\vec{v}_1\|^2 = 1$

$$\vec{v}_1 = (0, 0, -1) \quad \|\vec{v}_1\|^2 = 0 + 0 + 1 = 1$$

$$\vec{v}_2 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right) \quad \|\vec{v}_2\|^2 = \frac{1}{2} + \frac{1}{2} + 0 = 1$$

$$\vec{v}_3 = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right) \quad \|\vec{v}_3\|^2 = \frac{1}{2} + \frac{1}{2} + 0 = 1$$

$$\vec{v}_1 \cdot \vec{v}_2 = 0 + 0 + 0 = 0$$

$$\vec{v}_1 \cdot \vec{v}_3 = 0 + 0 + 0 = 0$$

$$\vec{v}_2 \cdot \vec{v}_3 = \frac{1}{2} + \left(-\frac{1}{2}\right) + 0 = 0$$

orthogonal

$$\begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -1 & 0 & 0 \end{bmatrix} [\vec{x}]_B = \begin{bmatrix} 5 \\ 2 \\ -3 \end{bmatrix}$$

$$[\vec{x}]_B = \begin{bmatrix} 0 + 0 + 3 \\ \frac{5}{\sqrt{2}} + \frac{2}{\sqrt{2}} + 0 \\ \frac{5}{\sqrt{2}} - \frac{2}{\sqrt{2}} + 0 \end{bmatrix} = \begin{bmatrix} 3 \\ \frac{7}{\sqrt{2}} \\ \frac{3}{\sqrt{2}} \end{bmatrix}$$

Projection onto an orthonormal basis

$$\begin{aligned}\text{proj}_V \vec{x} &= A(A^T A)^{-1} A^T \vec{x} \\ &= A I A^T \vec{x} \\ &= A A^T \vec{x}\end{aligned}$$

$$\left. \begin{aligned}\vec{v}_1 &= (0, 0, 1) \\ \vec{v}_2 &= \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)\end{aligned} \right\} V \quad \vec{x} = (1, 2, 3)$$

$$A = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} \\ 1 & 0 \end{bmatrix} \quad A^T = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

$$\begin{aligned}A A^T &= \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 + \frac{1}{2} & 0 + \frac{1}{2} & 0 + 0 \\ 0 + \frac{1}{2} & 0 + \frac{1}{2} & 0 + 0 \\ 0 + 0 & 0 + 0 & 1 + 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} + \frac{2}{2} + 0 \\ \frac{1}{2} + \frac{2}{2} + 0 \\ 0 + 0 + 3 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ \frac{3}{2} \\ 3 \end{bmatrix} = \text{proj}_V \vec{x}\end{aligned}$$

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Now, let's look at the product $C = A^T A$. The entry in the i -th row and j -th column of the resulting matrix is the dot product of the i -th row of A^T and the j -th column of A :

$$C_{ij} = \mathbf{v}_i^T \mathbf{v}_j$$

3. Apply the Orthonormal Properties

By definition, a set of vectors is **orthonormal** if it satisfies two specific conditions:

1. **Orthogonal:** Every vector is perpendicular to every other vector. The dot product of two different vectors is 0.
 - $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ when $i \neq j$.
2. **Normalized:** Every vector has a magnitude (length) of 1. The dot product of a vector with itself is 1.
 - $\mathbf{v}_i \cdot \mathbf{v}_i = 1$.

In mathematical terms, we use the Kronecker delta (δ_{ij}):

$$\mathbf{v}_i^T \mathbf{v}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Gram-Schmidt process for change of basis

$$V = \text{span}(\vec{v}_1, \vec{v}_2, \vec{v}_3)$$

$$= \text{span}(\vec{u}_1, \vec{v}_2, \vec{v}_3)$$

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$$\vec{u}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|}$$

$$\vec{w}_2 = \vec{v}_2 - (\vec{v}_2 \cdot \vec{u}_1) \vec{u}_1$$

$$\vec{w}_3 = \vec{v}_3 - [(\vec{v}_3 \cdot \vec{u}_1) \vec{u}_1 + (\vec{v}_3 \cdot \vec{u}_2) \vec{u}_2]$$

$\vdots$

$$\vec{w}_4 = \vec{v}_4 - [(\vec{v}_4 \cdot \vec{u}_1) \vec{u}_1 + (\vec{v}_4 \cdot \vec{u}_2) \vec{u}_2 + (\vec{v}_4 \cdot \vec{u}_3) \vec{u}_3]$$

Gram-Schmidt process for change of basis

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⋮

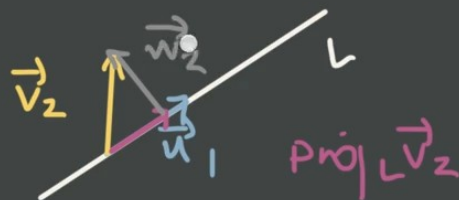
$$\vec{w}_4 = \vec{v}_4 - [(\vec{v}_4 \cdot \vec{u}_1) \vec{u}_1 + (\vec{v}_4 \cdot \vec{u}_2) \vec{u}_2 + (\vec{v}_4 \cdot \vec{u}_3) \vec{u}_3]$$

$$\vec{w}_2 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} - \left(\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \right) \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} - \left(-\frac{2}{\sqrt{2}} + 0 + 0 \right) \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} + \frac{2}{\sqrt{2}} \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$V = \text{span} \left(\begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ -1/\sqrt{3} \end{bmatrix} \right)$$



$$V = \text{span} \left(\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \right)$$

$$\|\vec{v}_1\| = \sqrt{1+1+0} = \sqrt{2}$$

$$\vec{u}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \quad \vec{u}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\|\vec{w}_2\| = \sqrt{1+1+1} = \sqrt{3}$$